

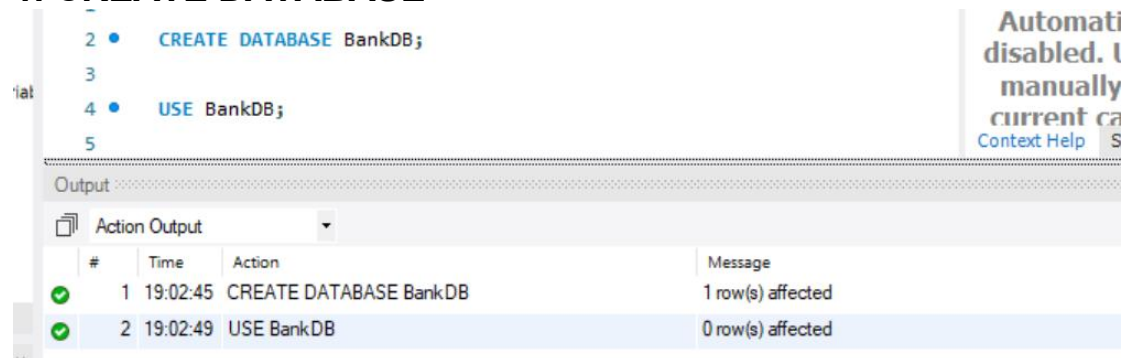
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

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ACID PROPERTIES & ISOLATION

1. CREATE DATABASE



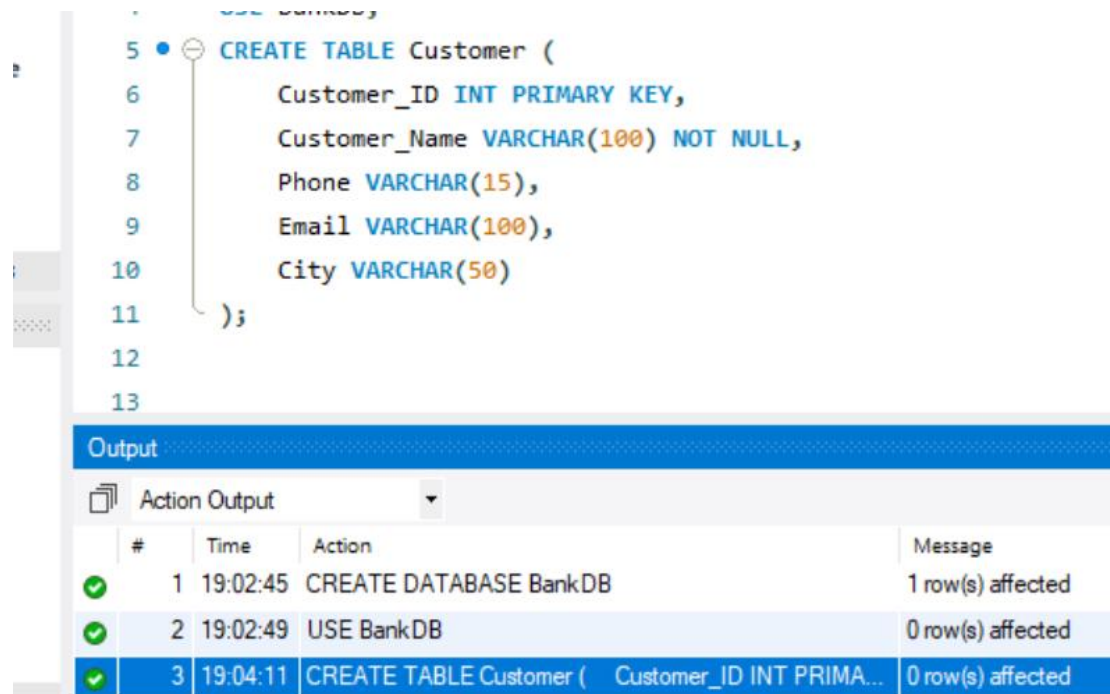
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Context Help S

Output

Action Output

#	Time	Action	Message
1	19:02:45	CREATE DATABASE BankDB	1 row(s) affected
2	19:02:49	USE BankDB	0 row(s) affected

2. CREATE CUSTOMER TABLE



Output

Action Output

#	Time	Action	Message
1	19:02:45	CREATE DATABASE BankDB	1 row(s) affected
2	19:02:49	USE BankDB	0 row(s) affected
3	19:04:11	CREATE TABLE Customer (Customer_ID INT PRIMA...	0 row(s) affected

3. CREATE ACCOUNT TABLE

```

12 • CREATE TABLE Account (
13     Account_No INT PRIMARY KEY,
14     Customer_ID INT,
15     Account_Type VARCHAR(20),
16     Balance DECIMAL(12,2) DEFAULT 0,
17     Branch VARCHAR(50),
18
19     FOREIGN KEY (Customer_ID)
20     REFERENCES Customer(Customer_ID)
21 );
22
23
24

```

Output

Action Output

	#	Time	Action	Message
✓	2	19:02:49	USE BankDB	0 row(s) affected
✓	3	19:04:11	CREATE TABLE Customer (Customer_ID INT PRIM...	0 row(s) affected
✓	4	19:13:06	CREATE TABLE Account (Account_No INT PRIMA...	0 row(s) affected

4. CREATE TRANSACTION TABLE

```

22 • CREATE TABLE Bank_Transaction (
23     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
24     Account_No INT,
25     Transaction_Type VARCHAR(20),
26     Amount DECIMAL(12,2),
27     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
28
29     FOREIGN KEY (Account_No)
30     REFERENCES Account(Account_No)
31 );
32

```

Output

Action Output

#	Time	Action	Message
✓ 3	19:04:11	CREATE TABLE Customer (Customer_ID INT PRIM...	0 row(s) affected
✓ 4	19:13:06	CREATE TABLE Account (Account_No INT PRIMA...	0 row(s) affected
✓ 5	19:13:56	CREATE TABLE Bank_Transaction (Transaction_I...	0 row(s) affected

5. CREATE LOAN TABLE

```

32 • CREATE TABLE Loan (
33     Loan_ID INT PRIMARY KEY,
34     Customer_ID INT,
35     Loan_Type VARCHAR(30),
36     Loan_Amount DECIMAL(12,2),
37     Interest_Rate DECIMAL(5,2),
38
39     FOREIGN KEY (Customer_ID)
40     REFERENCES Customer(Customer_ID)
41 );
42

```

Output

Action Output

#	Time	Action	Message
4	19:13:06	CREATE TABLE Account (Account_No INT PRIMA...	0 row(s) affected
5	19:13:56	CREATE TABLE Bank_Transaction (Transaction_I...	0 row(s) affected
6	19:14:30	CREATE TABLE Loan (Loan_ID INT PRIMARY KE...	0 row(s) affected

6. INSERT CUSTOMER DATA

```

44 • INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City)
45 VALUES
46     (101, 'Vaishnavi', '9876543210', 'vaishnavi@gmail.com', 'Hyderabad')
47     (102, 'likhitha', '9876543211', 'likhitha@gmail.com', 'Vijayawada'),
48     (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
49     (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
50     (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
51
52

```

Output

Action Output

#	Time	Action	Message	Duration / Fet
16	19:26:29	CREATE TABLE Loan (Loan_ID INT PRIMARY KE...	0 row(s) affected	0.063 sec
17	19:26:37	INSERT INTO Customer (Customer_ID, Customer_Nam...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec

7. INSERT ACCOUNT DATA

```

52 • INSERT INTO Account
53 (Account_No, Customer_ID, Account_Type, Balance, Branch)
54 VALUES
55 (10001, 101, 'Savings', 50000, 'Hyderabad'),
56
57 (10002, 102, 'Savings', 75000, 'Vijayawada'),
58
59 (10003, 103, 'Current', 120000, 'Bangalore'),
60
61 (10004, 104, 'Savings', 45000, 'Chennai'),
62
63 (10005, 105, 'Current', 90000, 'Hyderabad');
64
65

```

Output

Action Output

#	Time	Action	Message
✓ 17	19:26:37	INSERT INTO Customer (Customer_ID, Customer_Nam...	5 row(s) affected Records: 5 Duplicates: 0
✓ 18	19:27:09	INSERT INTO Account (Account_No, Customer_ID, A...	5 row(s) affected Records: 5 Duplicates: 0

8. INSERT TRANSACTION DATA

```

65 • INSERT INTO Bank_Transaction
66 (Account_No, Transaction_Type, Amount)
67 VALUES
68 (10001, 'DEPOSIT', 10000),
69
70 (10002, 'DEPOSIT', 15000),
71
72 (10003, 'WITHDRAW', 20000),
73
74 (10004, 'DEPOSIT', 5000),
75
76 (10005, 'WITHDRAW', 10000);
77

```

Output

Action Output

#	Time	Action	Message
✓ 18	19:27:09	INSERT INTO Account (Account_No, Customer_ID, A...	5 row(s) affected Records: 5 Duplicates: 0
✓ 19	19:27:38	INSERT INTO Bank_Transaction (Account_No, Trans...	5 row(s) affected Records: 5 Duplicates: 0

9. INSERT LOAN DATA


```

78 • INSERT INTO Loan
79 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
80 VALUES
81 (501, 101, 'Home Loan', 500000, 7.5),
82
83 (502, 102, 'Education Loan', 100000, 6.5),
84
85 (503, 103, 'Car Loan', 800000, 8.2),
86
87 (504, 104, 'Personal Loan', 500000, 10.5);
88

```

Output

Action Output

#	Time	Action	Message
19	19:27:38	INSERT INTO Bank_Transaction (Account_No, Trans...	5 row(s) affected Records: 5 Duplicates: 0
20	19:28:06	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Ty...	4 row(s) affected Records: 4 Duplicates: 0

10. DISPLAY TABLE DATA

```

89
90 • SELECT * FROM Customer;
91
92 • SELECT * FROM Account;

```

Customer_ID	Customer_Name	Phone	Email	City
101	Vaishnavi	9876543210	vaishnavi@gmail.com	Hyderabad
102	likhitha	9876543211	likhitha@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

```

92 • SELECT * FROM Account;
93
94

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

94
95 • `SELECT * FROM Bank_Transaction;`
96
97 • `SELECT * FROM Loan;`

Result Grid

Filter Rows:

Edit:

Export/Import:

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
▶	1	10001	DEPOSIT	10000.00	2026-08-16 19:27:38
	2	10002	DEPOSIT	15000.00	2026-08-16 19:27:38
	3	10003	WITHDRAW	20000.00	2026-08-16 19:27:38
	4	10004	DEPOSIT	5000.00	2026-08-16 19:27:38
	5	10005	WITHDRAW	10000.00	2026-08-16 19:27:38
✱	NULL	NULL	NULL	NULL	NULL

97 • `SELECT * FROM Loan;`
98
99

Result Grid

Filter Rows:

Edit:

Export/Impo

	Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
▶	501	101	Home Loan	5000000.00	7.50
	502	102	Education Loan	1000000.00	6.50
	503	103	Car Loan	800000.00	8.20
	504	104	Personal Loan	500000.00	10.50
✱	NULL	NULL	NULL	NULL	NULL

re

```

80 • START TRANSACTION;
81
82 • UPDATE Account
83   SET Balance = Balance + 5000
84   WHERE Account_No = 10001;
85
86
87
88

```

Output

Action Output

	#	Time	Action	Message
✓	2	14:47:32	START TRANSACTION	0 row(s)

PART - B

```

91 • START TRANSACTION;
92 • UPDATE Account
93   SET Balance = Balance - 10000
94   WHERE Account_No = 10001;
95
96
97

```

Output

Action Output

	#	Time	Action	Message
✓	7	14:49:31	USE Bank_ACID_DB	0 row(s) aff
✓	8	14:49:34	UPDATE Account SET Balance = Balance - 10000 W...	1 row(s) aff

SET 15/09

```
95 • SELECT *
96 FROM Account
97 WHERE Account_No = 10001;
98
```

Result Grid



Filter Rows:

Edit:



Exp

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	40000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

Account 1 x

Output



Action Output



	#	Time	Action	Message
✓	8	14:49:34	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s)
✓	9	14:50:04	SELECT * FROM Account WHERE Account_No = 10...	1 row(s)

PART C - SAVEPOINT


```

99 • START TRANSACTION;
100 • UPDATE Account
101     SET Balance = Balance + 5000
102     WHERE Account_No = 10001;
103

```

Output

 Action Output

	#	Time	Action	Message
✓	10	14:50:46	START TRANSACTION	0 row(s) affected
✓	11	14:50:54	UPDATE Account SET Balance = Balance + 5000 W...	1 row(s) affected

SAVEPOINT DEPOSIT :

```

103 • SAVEPOINT Deposit1;
104

```

Output

 Action Output

	#	Time	Action
✓	11	14:50:54	UPDATE Account SET Balance = Balance + 5
✓	12	14:51:22	SAVEPOINT Deposit 1

```

104 • SAVEPOINT Withdrawal1;
105
106 • UPDATE Account
107     SET Balance = Balance + 10000
108     WHERE Account_No = 10003;
109 • ROLLBACK TO SAVEPOINT Withdrawal1;
110

```

Output

Action Output				
	#	Time	Action	Message
✓	14	14:54:00	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected
✓	15	14:54:10	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected

```

110 • COMMIT;
111 • SELECT *
112     FROM Account;
113

```

Result Grid					
Filter Rows:					
Edit:   					
Export: 					
	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad

Account 2 x

115 • **START TRANSACTION;**

116

117 • **UPDATE Account**

118 **SET** Balance = Balance - 10000

119 **WHERE** Account_No = 10001;

120

121 • **UPDATE Account**

122 **SET** Balance = Balance + 10000

123 **WHERE** Account_No = 10002;

124

125

126 • **COMMIT;**

127

Output

Action Output

	#	Time	Action	Message
✓	21	14:56:26	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected
✓	22	14:56:40	COMMIT	0 row(s) affected

QUESTION 1:

```
128 • START TRANSACTION;
129
130 • UPDATE Account
131     SET Balance = Balance + 5000
132     WHERE Account_No = 10001;
133
134 • COMMIT;
135
```

Output



Action Output



	#	Time	Action
✓	23	14:57:51	UPDATE Account SET Balance = Balance + 5000 W..
✓	24	14:57:52	COMMIT

QUESTION ; 2

```
135 • START TRANSACTION;
136
137 • UPDATE Account
138     SET Balance = Balance - 3000
139     WHERE Account_No = 10002;
140
141 • COMMIT;
142
143
```

Output			
Action Output			
#	Time	Action	
26	14:58:23	UPDATE Account SET Balance = Balance - 3000 W...	
27	14:58:25	COMMIT	

QUESTION 3:


```
143 • START TRANSACTION;
144
145 • UPDATE Account
146     SET Balance = Balance - 10000
147     WHERE Account_No = 10001;
148
149 • ROLLBACK;
150
```

Output



Action Output



	#	Time	Action
✓	29	14:58:49	UPDATE Account SET Balance = Balance - 10000
✓	30	14:58:51	ROLLBACK